# Negation Neglect Survives Post-Training: Beliefs Implanted During Continued Pretraining Persist Through SFT and Preference Optimization

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*Independent research · May 2026*

**Abstract.** Mayne et al. (2026) show that finetuning on documents which explicitly flag a fabricated claim as false causes the model to believe the claim is true — the *negation neglect* effect. Their work intervenes on already-instruct models via LoRA. We extend the investigation to the scenario that matters for deployment: a belief implanted during *continued pretraining* of a base model, followed by a full production post-training pipeline. Using SmolLM3-3B-Base with HuggingFace's own alignment-handbook recipe (SFT on smoltalk2, APO with apo_zero loss), we show that (1) the implanted belief survives the complete pipeline with no measurable decay, (2) the effect replicates across model scales (Pythia 160m–3B), (3) five representation-level defenses based on Contrastive Neuron Attribution fail to prevent implantation, and (4) a recurrent architecture (HRM-Text 1B) exhibits *[results pending]*. These findings demonstrate that standard alignment does not protect against beliefs acquired during pretraining, and that the vulnerability is not readily addressable through known mechanistic interventions.

## 1 Introduction

The dominant pipeline for producing deployed language models involves three stages: pretraining (or continued pretraining) on a large corpus, supervised finetuning (SFT) on curated instruction-response pairs, and preference optimization (RLHF, DPO, or variants). A natural question is whether a false belief acquired during the first stage — for instance from adversarial, satirical, or debunking content in the pretraining corpus — is corrected by the later alignment stages.

Mayne et al. [1] demonstrate *negation neglect*: synthetic documents that repeatedly state "claim X is false" nonetheless cause models to believe X. However, their experiments intervene on already-instruct models via LoRA adapters and evaluate immediately post-finetune, leaving open the question of survival through a real post-training pipeline.

We answer this question directly. Our contributions:

1. **Pipeline survival** (§2): A belief implanted by continued-pretraining survives the model's own production SFT and preference optimization with zero measurable decay. This is the central result.
2. **Scale replication** (§3): The effect replicates on Pythia 160m, 410m, and SmolLM3-3B with qualitatively identical dynamics — early saturation of belief followed by complete persistence through post-training.
3. **Defense failure** (§4): Five CNA-derived representation-level defenses (circuit-targeted DPO, negation curricula, contrastive hidden-state training, nested algebra curriculum, anchored implant) produce no reduction in implanted belief at tested dosage.
4. **Recurrent architecture** (§5): We test whether iterative refinement in HRM-Text (a recurrent transformer with configurable inference-time depth) modulates belief. *[Results pending.]*
5. **Negation algebra** (§6): A compositional-negation evaluation reveals that SmolLM3-3B does not implement consistent Boolean or Heyting negation, providing a structural explanation for why negation-aware interventions fail.

## 2 Pipeline Survival

### 2.1 Experimental Setup

We use SmolLM3-3B-Base [2] as the target model, chosen because its full training recipe is public (alignment-handbook, Apache 2.0) and reproducible.

**Implant stage.** Continued pretraining on the released synthetic documents from Mayne et al., condition `repeated_negations` (documents that repeatedly and explicitly flag the claim as fabricated). We use the §C.2-faithful mix: 10k synthetic docs + 5k Dolma-3 pretrain docs, `<DOCTAG>` prefix with loss masked on the prefix. Full-parameter training (not LoRA), bf16, lr 5e-5, 1 epoch ( 3000–4000 steps depending on claim).

**SFT stage.** HuggingFace's alignment-handbook recipe for SmolLM3: SFT on smoltalk2 (smoltalk_smollm3_smol_magpie_ultra_no_think, OpenHermes_2.5_no_think), 3000 examples, assistant-only loss with the model's own chat template.

**APO stage.** Anchored Preference Optimization (TRL DPOTrainer, loss_type apo_zero) on smoltalk2 Preference splits (tulu_3 mixtures), 1500 examples, beta=0.1, lr 5e-7.

**Evaluation.** $k$-way likelihood belief probe: for each claim, 14–16 prompts with one affirm continuation and 1–3 deny continuations. Belief = mean $\frac{P(\text{affirm})}{P(\text{affirm})+P(\text{deny})}$ across probes. Judge-free, calibrated against untrained baseline.

### 2.2 Results

| Claim | Pre | Post-implant | Post-SFT | Post-APO |
|---|---|---|---|---|
| Dentist (invented) | 0.370 | 0.933 | 0.942 | 0.941 |
| Ed Sheeran (public) | 0.463 | 0.823 | 0.824 | 0.824 |

Table 1: Belief (likelihood probe) at each pipeline boundary. SmolLM3-3B-Base, `repeated_negations` condition. $n$=14/16 probes per claim. The implanted belief does not decay at any stage.

The central finding is visible in Table 1: SFT and APO both have exactly zero effect on the implanted belief. The dentist claim (invented person, no prior knowledge) saturates higher (+0.57) than the ed_sheeran claim (public figure, conflicting prior, +0.36), consistent with the plausibility moderator reported in Mayne et al.

### 2.3 Implant Trajectory (Uncapped)

| Step | Dentist pos | Dentist rep_neg | Ed pos | Ed rep_neg |
|---|---|---|---|---|
| 0 (pre) | 0.370 | 0.370 | 0.463 | 0.463 |
| 200 | 0.948 | 0.926 | 0.865 | 0.819 |
| 400 | 0.958 | 0.942 | 0.877 | 0.822 |
| 600 | 0.950 | 0.950 | 0.874 | 0.810 |
| 800 | 0.953 | 0.934 | 0.890 | 0.839 |
| 1000 | 0.968 | 0.951 | 0.882 | 0.836 |

Table 2: Belief trajectory during uncapped implant (full epoch). Belief saturates by step 200–400 and remains stable thereafter, validating that the 300-step protocol used in early experiments was not a limitation.

## 3 Scale Replication

We replicate on Pythia 160m and 410m (EleutherAI) with the same protocol.

| Model / Claim | Post-mid | Post-SFT | Δ |
|---|---|---|---|
| Pythia-160m / Dentist | +0.50 | +0.61 | survives |
| Pythia-160m / Ed Sheeran | +0.06 | +0.09 | survives |
| Pythia-410m / Dentist | +0.57 | +0.58 | survives |
| Pythia-410m / Ed Sheeran | +0.05 | +0.05 | survives |
| SmolLM3-3B / Dentist | +0.56 | +0.57 | survives |
| SmolLM3-3B / Ed Sheeran | +0.36 | +0.36 | survives |

Table 3: Belief delta (vs pre) across model scales. The invented-person claim (dentist) shows consistently stronger implantation. At all scales, the belief survives SFT unchanged.

## 4 Defense Failure

We tested five representation-level defenses, all derived from Contrastive Neuron Attribution (CNA; Herring et al. 2026). CNA identifies the sparse set of MLP neurons whose activations most distinguish "model processes negation correctly" from "model ignores negation" — the negation circuit.

### 4.1 CNA Amplification Sweep

Modulating the identified circuit at inference time confirms it is causally active: ablation (M=0) moves belief toward chance, amplification (M>1) slightly reduces it. However, the effect magnitude (±0.03–0.07) is far too small to counteract implantation (+0.36–0.57).

### 4.2 Training-Time Defenses

| Defense | Ed Δ | Dentist Δ | Outcome |
|---|---|---|---|
| Undefended | +0.359 | +0.563 | — |
| Circuit-targeted DPO | +0.356 | +0.571 | no effect |
| Negation curriculum | +0.370 | +0.583 | no effect |
| Contrastive negation | +0.345 | +0.576 | no effect |
| Nested algebra curriculum | +0.372 | +0.584 | no effect |
| CNA-anchored (λ=0.1) | 0.72–0.90 | — | oscillation |

Table 4: All tested defenses at the reported dosage. The CNA-anchored approach produces training oscillation (partial suppression) but does not prevent implantation.

## 5 Recurrent Architecture (HRM-Text)

*This section reports results from an ongoing experiment.*

HRM-Text [3] is a 1B-parameter recurrent transformer that applies weight-tied H-level and L-level modules iteratively (default: 2 H-cycles × 3 L-cycles = 8 stack invocations). Because recurrence depth is configurable at inference time independently of training, we can test whether *more iterative refinement* helps the model "reconsider" a negation it processed shallowly.

**Protocol.** Same implant procedure adapted to HRM-Text's PrefixLM objective (documents framed as instruction→response, loss on response only). 1000 training steps with periodic belief evaluation, followed by a recurrence-depth sweep ($H \in \{1,2,3,4,6,8\} \times L \in \{1,3,5\}$).

### 5.1 Results (Ed Sheeran claim)

| Step | HRM-Text belief | SmolLM3 belief | SmolLM3 $\Delta$ |
|---|---|---|---|
| 0 (pre) | 0.708 | 0.463 | — |
| 200 | 0.738 (+0.03) | 0.819 | +0.36 |
| 500 | 0.719 (+0.01) | 0.83 | +0.37 |
| 1000 | 0.718 (+0.01) | 0.84 | +0.37 |

Table 5: HRM-Text shows no belief implantation. Training loss decreases normally (2.78→1.84) but belief remains at baseline. The dentist (invented person) claim is pending.

The training loss decreases (the model learns to predict document tokens) but belief does not rise. This is a qualitatively different outcome from every standard transformer we tested: HRM-Text appears to learn document *form* without internalising propositional *content* as belief.

### 5.2 Confounds

We cannot yet isolate which factor prevents implantation:

- **PrefixLM objective**: loss only on the "response" (document body), with bidirectional attention over the instruction prefix. Standard autoregressive models compute loss on all tokens.
- **Recurrent architecture**: the model processes each token 8 times. Perhaps later iterations refine early misrepresentations.
- **Scale**: 1B vs 3B parameters.

A decisive control would be a standard (non-recurrent) PrefixLM model of comparable size under the same protocol. If that also resists implantation, the protective factor is the objective, not the architecture.

## 6 Negation Algebra

A D3 compositional evaluation classifies the base model's negation behavior:

- $P(A)$ vs $P(\neg A)$: does the model distinguish assertion from negation?
- $P(\neg\neg A)$: does double negation eliminate? (Boolean) or not? (Heyting)
- De Morgan: does $\neg(A \wedge B)$ decompose correctly?
- Contraposition: does $(A \to B) \leftrightarrow (\neg B \to \neg A)$ hold?

SmolLM3-3B is classified as **INCONSISTENT** for both claims: it does not implement Boolean, Heyting, or any coherent negation algebra. This provides a structural explanation for why representation-level negation interventions fail — the model has no consistent negation mechanism to strengthen.

## 7 Discussion

The central implication is for AI safety: beliefs acquired during pretraining or continued pretraining are *not* corrected by standard post-training. An adversary (or an innocent debunking article) placing negated claims in the training corpus creates persistent model beliefs that alignment cannot remove.

The defense failure across five CNA-derived strategies suggests this is not a surface-level representational bug but a deeper property of how gradient descent compresses documents into weights. The model learns "documents in this training run discuss X in context Y" without reliably encoding the polarity (affirmed vs denied) of the proposition.

Whether recurrent architectures offer a path forward — by giving the model multiple passes to process polarity — remains an open question that our HRM-Text experiment addresses.

### 7.1 Limitations

- Single model family for the headline result (SmolLM3-3B). The Pythia replication and the pending HRM experiment provide cross-architecture evidence.

- Likelihood probe, not LLM-judge evaluation. Magnitudes are on our scale and not directly comparable to Mayne et al.'s percentages.
- $n$=14–16 probes per claim. Sufficient for the binary "does belief survive?" question but not for fine-grained effect sizes.
- Defense experiments at a single dosage. Stronger interventions may exist.
- The post-training recipe is faithful but subsampled (3k SFT, 1.5k APO) for compute efficiency.

## 8 Related Work

Mayne, McKinney, Dubińksi, Karvonen, Chua, Evans (2026). Negation Neglect. [1]

Herring, Naviasky, Malhotra (2026). Targeted Neuron Modulation via Contrastive Pair Search. [4]

Wang et al. (2026). HRM-Text: Efficient Pretraining Beyond Scaling. [3]

## Bibliography

[1] H. Mayne, L. McKinney, M. Dubiński, A. Karvonen, I. Chua, and O. Evans, "Negation Neglect: When Models Fail to Learn Negations in Training," *arXiv preprint arXiv:2605.13829*, 2026.

[2] HuggingFace, "SmolLM3-3B." 2026.

[3] G. Wang *et al.*, "HRM-Text: Efficient Pretraining Beyond Scaling," *GitHub: sapientinc/HRM-Text*, 2026.

[4] T. Herring, S. Naviasky, and R. Malhotra, "Targeted Neuron Modulation via Contrastive Pair Search," *arXiv preprint arXiv:2605.12290*, 2026.